Summary of the revision

The authors have substantially revised the paper. The most consequential changes are: (i) a second wave of data collection that almost doubles the sample (105 $\rightarrow$ 207) and expands the set of information structures from 7 to 12 per prior (48 treatments total); (ii) a rewritten theory section that removes the old Proposition 2 (imprudence) and replaces the first-order Taylor argument in the old Proposition 3 with an exact proof; (iii) a reorganised empirical section that frames Section 5.2 around deviations from the risk-neutral, Bayesian benchmark and consolidates the discussion around alternative mechanisms (risk preferences, probability weighting, anchoring, non-instrumental information, anticipation of IP bias, and failure to distinguish FP/FN) in the new Section 5.3; and (iv) additional empirical content including: statistical tests for equality of FP and FN coefficients, confidence intervals and a regression equation for Figure 5, a clarified treatment assignment protocol, clarified posterior definitions, significance stars, expanded literature, and expanded policy discussion in the conclusion.

Assessment

Overall, I appreciate the authors' effort in the revision (especially the extra sessions). Below I list a couple of comments, mostly about clarification, framing and the handling of alternative explanations.

Comments

1. Treatments and Treatment Assignment:

   (a) While the new "Treatments and Treatment Assignment" subsection on page 10 provides additional detail on how treatments are assigned, it remains incomplete. It states that each subject is randomly allocated to a pair of priors but does not explain how information structures are assigned.

   (b) The text also notes that "the first treatment was always the one with a fully truthful information structure," which suggests that information structures are not randomly assigned across subjects. For transparency and to avoid confusion, the authors should provide at least a general description and justification of how information structures are allocated.

(c) Because information structures do not appear to be randomly assigned, the resulting sample sizes across treatment cells (i.e., prior $\times$ information structure combinations) are likely to differ. I encourage the authors to report sample sizes by treatment cell. Given the large number of cells ($4 \times 12 = 48$), a table in the appendix would likely suffice.

2. Description and groupings in Tables 2, 3, and A4:

   (a) On page 12, the text states that "Table 2 presents the average protection decisions by prior and signal type," but Table 2 does not, in fact, report descriptive statistics by prior. Instead, it groups observations by signal value (white/black) and by information-structure type (truthful, FP-only, FN-only, both errors). The same mismatch appears in the caption of Appendix Table A4, which refers to "signal type and prior," although the table is actually organized by signal value and information-structure type.

   (b) Relatedly, the authors could briefly justify why Table 2 groups observations by signal value and information structure while pooling across priors. A single sentence would suffice. Providing this explanation would improve transparency and clarity for readers, especially since the same grouping is used in Tables 3 and A4.

3. Claim on page 16 regarding the introduction of the FN rate (row 1 vs. 2):

   (a) The paragraph below Table 3 states that introducing the FP rate (row 1 vs. 3 of Table 3) magnifies the upward updating bias for white signals—which is correct: the Bayesian posterior remains at 0, while the mean updating error increases from 0.048 to 0.210. The text then claims that "we find a similar effect for the introduction of the FN rate (row 1 vs. 2)." However, comparing row 1 and row 2, the mean updating error does not appear to change significantly (0.048 vs. 0.051).

   (b) Moreover, introducing the FN rate changes the Bayesian posterior for the white signal (from 0.000 to 0.119), so the comparison between row 1 and row 2 does not hold the posterior fixed—unlike the comparison between row 1 and row 3.

4. Section 4's closing summary understates the evidence presented in Tables 2 and 3:

   (a) The paragraph just before Section 5 (page 17) concludes that "All of this suggests that subjects understand the experimental environment well enough to allow us to draw reasonable inferences from our WTP elicitation." While I agree that subjects

appear to understand the environment sufficiently well, this framing understates several important findings from Section 4.

(b) For example, Tables 2 and 3 show that increasing the FP (FN) rate affects protection decisions and updating errors even in cases where the rate should not influence the Bayesian posterior (e.g., FP for a white signal in Table 3, row 1 vs. 3). These patterns provide suggestive evidence of the FP/FN-confusion mechanism. I would encourage the authors to add a sentence foreshadowing that the descriptive analysis already reveals patterns that are later revisited and explained in Section 5.3. This would make the main mechanism visible from Section 4 onward, rather than only at the end of the paper.

5. Hypothesis 1 and Result 1 overstate what the t-tests in Table 4 establish:

   (a) Hypothesis 1 (page 17) states that "subjects' WTPs for signals are equal to their value for risk-neutral agents," and Result 1 concludes that "on average, there are no significant discrepancies between the empirical WTP and the risk-neutral benchmark." However, the supporting analysis relies on a t-test of the mean of $\Delta b_{is} = b_{is} - b^* s$. Failing to reject $E[\Delta bis] = 0$ is a much weaker result than establishing $\Delta b_{is} = 0$, as large but offsetting over- and under-valuations across subjects and treatments may cancel out in the mean. Given the likely presence of individual heterogeneity, this distinction is important. I suggest one of the following revisions:

      i. Restate Hypothesis 1 and Result 1 more precisely, for example: "subjects' WTPs are not systematically above or below the risk-neutral benchmark on average," which is itself an informative finding; or

      ii. Test the stronger claim directly, for instance by examining whether the absolute deviation $|\Delta b_{is}|$ is significantly different from zero.

6. (Optional) Reorganize Section 5.3 to give greater prominence to the FP/FN-confusion mechanism:

   (a) Given that the FP/FN-confusion mechanism is the primary mechanism, it might be preferable to present it as the first candidate explanation rather than the last, and then proceed to evaluate and rule out alternative mechanisms.

7. In the risk-preference discussion (pp. 23–24), the authors could also note that the main effects of FP and FN costs remain significant, rather than focusing solely on the insignificance of the interaction terms.

8. Probability weighting discussion (pp. 24–25):

   (a) In equation 9 and the defintion of Λ, the numerator should be P_{01} instead of P_{00}?

   (b) I appreciate the authors' careful argument with rank-depedent utility! In my previous report, I was meant to refer to the probability weighting function from prospect theory (Kahneman and Tversky, 1979). But the arguments turn out to be quite similar because the particular payoff set used in the experiment.

   (c) The flow of the argument is clear, and the observation that "$\pi + (1 - \pi)P_{10} \leq 0.75$, meaning that the value is still likely to fall in the mid-range of an S-shaped weighting function," is insightful. However, this depends on the rotation point of the probability weighting function. Theoretically, if the rotation point lies far to the left, then $\pi + (1 - \pi)P_{10} \leq 0.75$ might still fall within the steeper region of the weighting function (such that the denominator is large and Λ is small, implying lower relative sensitivity). It would be helpful if the authors could provide evidence for "middle" rotation points in existing studies (e.g., Gonzalez and Wu (1999), as cited). That said, since the shape of the weighting function is not restricted, this possibility may be too flexible to rule out. In any case, it would be helpful to discuss this issue more thoroughly.

9. The "Anticipation of Bias in Protection Decision":

   (a) Table A10 should be Table A8; the caption of Table A8 should refer to WTP deviations rather than simply WTP.

   (b) More importantly, I am not entirely convinced that the two newly introduced variables adequately test the mechanism of "anticipation of bias in protection decisions." If I understand correctly, Δ^{prot} measures the extra cost incurred by the subject relative to the optimal benchmark, while Δ^{loss} measures the additional loss incurred relative to the same benchmark. The coefficients in Table A8 are negative, suggesting that greater mistakes (in terms of cost or loss) reduce WTP deviations. In other words, subjects who perform worse in the IP task appear to be more accurate in their WTP, which may indicate that they account for their own mistakes.

   (c) Given that the coefficients on the two new variables are statistically significant, the conclusion that "subjects' biases in how they will utilize signals when facing an IP task do not explain the observed pattern of sensitivities to FP and FN rates" seems premature. A more appropriate interpretation is that the coefficients on FP and FN do not appear to change after introducing these two variables.

(d) However, a more informative regression specification would include interaction terms between the mistake variables and FP/FN rates. The intuition is that, if anticipation of bias is the primary channel, then mistakes in protection decisions should mediate the relationship between WTP deviations and FP/FN. That is, if subjects expect to make more mistakes in the IP task at higher FP/FN rates, they may reduce their WTP, leading to larger deviations. Along similar lines, it may be preferable to distinguish mistakes in BP and IP directly, rather than separating them into cost and loss components.

(e) Finally, even if anticipation of bias is an important (or primary) channel, it does not undermine the FP/FN-confusion mechanism; if anything, it reinforces it. Tables 2 and 7 already show that subjects' IP protection decisions respond to FP and FN rates in ways consistent with FP/FN confusion. For instance, subjects choose to protect upon receiving a white signal even when there are no false negatives but only false positives (Table 7 provides related evidence). Anticipation of bias suggests that this confusion arises not only at the WTP stage but also during the IP stage, with such anticipation leading to larger deviations of WTP from the risk-neutral optimal benchmark.

10. The authors should consider incorporating the cross-task consistency analysis from the response letter (Table 2 on page 13) into the paper, or at least including it in the appendix.

11. (Optional) The "dishonest gremlin" framing: I understand the authors' argument that re-running the experiment with a different framing of the information structure might make pooling unjustifiable. However, the authors could consider adding a brief discussion of this issue (perhaps at the end of the paper). Even without the "dishonest" framing, individuals may still conflate the two types of errors (FP and FN). It would therefore be helpful for the authors to acknowledge and discuss this potential limitation.

12. Exposition:

    (a) On page 11, the sentence "this section also provides ... the theoretical predictions" is a tad imprecise: the predictions refer to those of the risk-neutral benchmark, not theoretical predictions in general. I would suggest "... from the theoretical predictions of the risk-neutral benchmark" (or similar) for clarity.

    (b) Relabel "Posterior" as "Bayesian posterior" in Tables 2, 3, A2, and A4. It would help the readers in case they mistaken it as elicted posterior.

(c) In both Table 5 and Table A8, the boundary between the main regression panel and the lower panel reporting the risk-averse and risk-loving subgroup tests is not easy to spot at a glance. A horizontal rule separating the two would make the structure of these tables much easier to spot.

(d) (Optional.) The appendix tables would be easier to navigate if the captions (or notes) pointed the reader to the coefficient or column that matters for the corresponding discussion in the main text. A short closing sentence in each catpion — e.g., "comparing the coefficient of XYZ and ABC in column 2 suggests that ..." — would substantially reduce the cost of cross-referencing.

(e) On page 21, the text refers to "the systematic WTP biases reported above" without specifying which biases are meant. I would suggest the authors to be more precise here

13. Typos:

(a) There is an extra "=" sign in equation (2) on page 5.

(b) On page 7, "Equations 6-s 7" should read "Equations 6 and 7" (or similar).

(c) Footnotes 10 and 11 are missing a full stop at the end.

(d) Footnote 13 (page 21): "average these errors across all the decisions with identical priors, false positive, and false negative rates". Should "identical" be "different"?

(e) In the last paragraph on page 26, Table A10 should be Table A8.